Shift in perspective : Flow profiles (microscopic look) → Large scale Info
- → Total force on wall
- → Overall flow rate
- → Total work performed by a pump/turbine mixer

Hard to model $v(r, \theta, z)$

Macroscopic Mass Balance

Macro / Micro / Meso

$v = v_i \hat{e}_s$

Mass balance on a CV : $\frac{dm_{CV}}{dt} + \iint_{CS} \rho (\hat{n} \cdot \underline{v}) \, dS = 0$

- → Multiple inlet & outlet : $\sum \rho_i \iint_{A_i} (\hat{n} \cdot \underline{v}) \, dS$
- → constant density

$\underline{v} = v \hat{e}_v$ (mag. × "velocity" direction [constant @ each inlet/outlet] (or unit vector))

$(\hat{n} \cdot \underline{v}) = v (\hat{n} \cdot \hat{e}_v) \Rightarrow v (\|\hat{n}\| \|\hat{e}_v\| \cos\theta)$

$\underline{a} \cdot \underline{b} = \|a\| \|b\| \cos\theta$ — dot product

$\sum \rho_i \iint_{A_i} v \cos\theta \, dS = \sum \rho_i \cos\theta \iint_{A_i} v \, dS$

$\langle v \rangle = \frac{\iint_{A_i} v \cdot dS}{\iint_{A_i} dS} = \frac{1}{A_i} \iint_{A_i} v \cdot dS$

$\therefore \iint_{A_i} v \cdot ds = \langle v \rangle A_i$

$\frac{dm_{CV}}{dt} + \sum_{i=1}^{\#\text{ streams}} [\rho A \cos\theta \langle v \rangle] = 0$

θ_i = Angle b/w $\hat{n}_i$ & $\hat{v}_i$ (will carry the negative sign depending on inlet & outlet)

Macroscopic Momentum Balance

$F = \frac{dP}{dt} = \frac{d(mv)}{dt}$

$\frac{dP}{dt} + \iint_{CS} (\hat{n} \cdot \underline{v}) \rho \underline{v} \, dS = \sum_{\text{on CV}} \underline{f}$

$\underline{v} = v \hat{e}_v$

$\sum \rho_i \cos\theta \, v \iint v^2 \cdot dS = \sum_{i=1}^{\#\text{ streams}} \rho A \cos\theta \langle v^2 \rangle \hat{v}$ (somewhat hard to compute)

$\langle v^2 \rangle = \frac{\iint v^2 \cdot d_s}{A_i}$

$\beta = 1$ (turbulent) ; $\beta = 0.75$ (laminar)

$\beta = \frac{\langle v^2 \rangle}{\langle v \rangle^2}$; $\beta = 1$, $\beta = 0.75$, $\langle v^2 \rangle$, $\langle v \rangle^2 \to \beta$

$\frac{dP}{dt} + \sum_{i=1}^{\#} \left[\frac{\rho A \cos\theta \langle v \rangle^2}{\beta} \hat{v} \right] = \sum_{\text{on CV}} \underline{f}$ (unit vector!)

$= \underline{f}_g + \underline{f}_{\text{inlet}} + \underline{f}_{\text{outlet}} + \underline{f}_{\text{walls}}$

mg

$\iint [\hat{n} \cdot \Pi] \, ds = \iint \hat{n} \cdot [-p\underline{I} + \underline{\tau}] \, ds$

$= \iint -p \hat{n} \, ds + \iint \hat{n} \cdot \underline{\underline{\tau}} \, ds$ (pressure dominated; $\underline{\underline{\tau}} \to 0$; viscous terms accounted for in $\underline{f}_{\text{walls}}$!)

$\underline{f}_{in} = -P_1 A_1 \hat{n}_{A_1}$

$\underline{f}_{out} = -P_2 A_2 \hat{n}_{A_2}$

Macroscopic Energy Balance

$\frac{d\hat{E}}{dt} + \iint_{CS} (\hat{n} \cdot \underline{v}) \rho \hat{E} \, dS = Q_{in} - W_{by}$

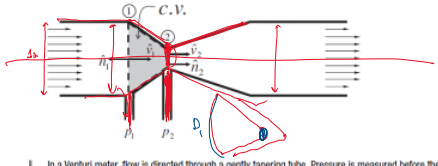
$\alpha = \frac{\langle v \rangle^2}{\langle v^2 \rangle}$; $\alpha \approx 1$ (turbulent)

- → Open sys,
- → $E = E_K + E_P + U$
- → constant velocity across inlet/outlet
- → $\hat{H} = \hat{U} + P\hat{V}$
- → Flow work / shaft work
- → No phase change
- → SS, single input/single output
- → $\Delta T = 0$, no reaction

$\frac{\Delta P}{\rho} + \frac{\Delta \langle v \rangle^2}{2\alpha} + g \Delta z + F = -\frac{W_{s,by}}{m}$

MEB

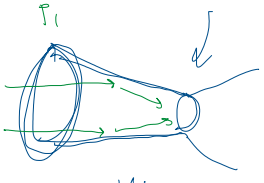
PROBLEM: 1.21 A Venturi meter with a 1.00 *in* diameter throat is to be installed into a 2 *in* line (Schedule 40 piping, smooth) with water flowing at 25°C. If the flow is turbulent and the range of expected flow rates is 0 – 200 gpm, what is the expected range of pressure drop in the Venturi meter? You may neglect friction.



In a Venturi meter, flow is directed through a gently tapering tube. Pressure is measured before the contraction (1) and at the point of smallest diameter (2) (i.e., the throat). The relationship between the measured pressures and the fluid velocity may be deduced from the mechanical energy balance with assistance of the macroscopic mass balance.

2" Schedule 40 pipe: ID = 2.067"

Pg 792 [Ex. 9.5]



(1) Velocity relationships (v_1 vs. v_2) : v_1, D_1 ; v_2, D_2 — Mass balance

balance.

(1) Velocity relationships (v_1 vs. v_2)? v_1, D_1 — v_2, D_2 } Mass balance

$$\frac{dm_{cv}}{dt}^{0} + \sum_i \rho_i A_i \cos\theta_i \langle v \rangle_i = 0$$

$$\rho_1 A_1 \cos\theta_1 \langle v_1 \rangle + \rho_2 A_2 \cos\theta_2 \langle v_2 \rangle = 0$$

$$\rho_1 \frac{\pi D_1^2}{4} (-1) \langle v_1 \rangle + \rho_2 A_2 (1) \langle v_2 \rangle = 0$$

$$\rho_1 \frac{\pi D_1^2}{4} \langle v_1 \rangle = \rho_2 \frac{\pi D_2^2}{4} \langle v_2 \rangle$$

$$Q_1 = Q_2 \ (@SS)$$

$$\boxed{\langle v_1 \rangle = \left(\frac{D_2}{D_1}\right)^2 \langle v_2 \rangle}$$

P_1 — $-\Delta P$ minimum

$D_1 \gg D_2$

$1 - 1 \to 0$

V_s

$$\frac{\Delta p}{\rho}^{\checkmark} + \Delta z^{0} + \frac{\Delta \langle v \rangle^2}{2\alpha} + F^{0} = -\frac{W_{s,by}}{m}^{0}$$

Ignore your $\langle v \rangle$

$$\frac{P_2 - P_1}{\rho} = -\frac{(\langle v_2 \rangle^2 - \langle v_1 \rangle^2)}{2} \quad ; \quad \alpha = 1 \ (\text{turbulent})$$

$$\text{Pressure drop} = -\Delta P = P_1 - P_2 = \frac{\rho \bar{V}_2^2}{2}\left(1 - \left(\frac{D_2}{D_1}\right)^4\right)$$

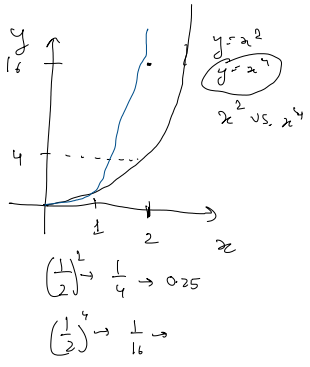
D_1^2 vs. D_2^4

$D_2 \uparrow\uparrow$

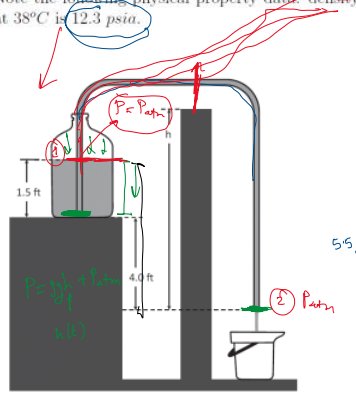
$\simeq 0$

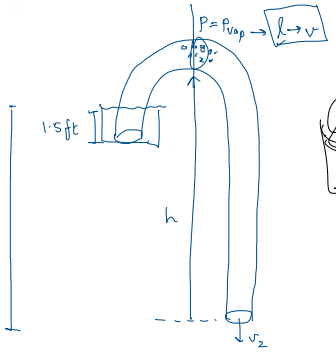
$P_1 - P_2 \simeq 0$

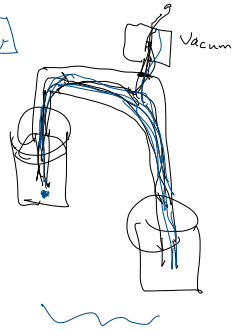
$P_1 = P_2$



PROBLEM: 1.23 A gasoline tank is connected to a 25 *ft* hose (ID = 1.50 *cm*) as shown in Figure 1.49. The ambient temperature is 38°C. What is the maximum height of the barrier over which the gasoline may be siphoned? You may neglect frictional losses. Note the following physical property data: density of gasoline=5.6 lb_m/gal, vapor pressure at 38°C is 12.3 *psia*.





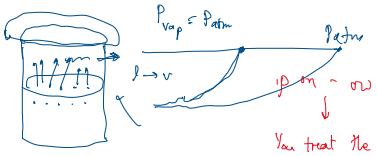


(1) What is the exit velocity?

$$\frac{\Delta P}{\rho} + \frac{\Delta \langle v \rangle^2}{2\alpha} + g\Delta z + F = \frac{W_s}{m}^{0}$$

$(z_2 - z_1)$

$$\langle v_2 \rangle^2 - \Delta P^{0} + g(66.5) = 0$$



You treat the fluid as a continuum (no breaks) (free to choose the pressure that is the most convinient)

↓

As liquid level is falling (the pressure @ bottom still be changing)

↓

Hard to use that

$\frac{P_1}{\rho} + \frac{v_1^2}{2\alpha} + gz_1 + h_f = \frac{v_2^2}{2\alpha}$

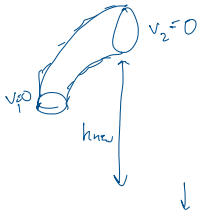
$\frac{v_2^2 - 0}{2} + g(66.5) = 0$

$\langle v_2 \rangle = \sqrt{-2g(-5.5)}$

$v_2 = ___$

② Max height: fluid will stop flowing when suction breaks (i.e. gas/air is introduced somehow)

↳ This can happen when the pressure equals vapor pressure (to "boil" the liquid)



$P_{vap} = 12.3$

$\frac{P_2 - P_1}{\rho} + \frac{v_2^2 - v_1^2}{2} + g(-h) = 0$

Solve for h

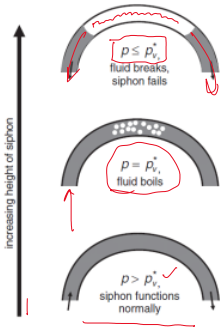
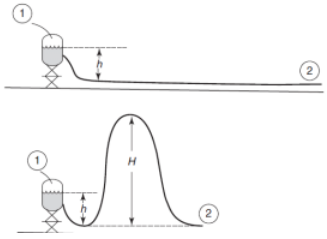


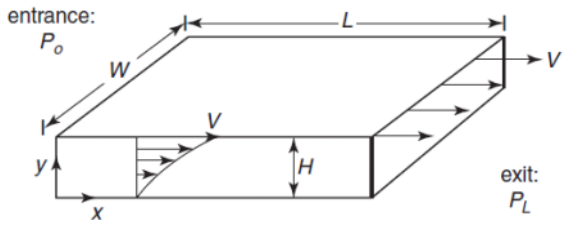
Figure 1.16 Schematic of the formation of a vapor gap in a siphon. This occurs when the barrier over which the fluid travels rises above a critical height; the critical height may be calculated as shown in the example.

PROBLEM: 1.24 A water tank is connected to a 100 *ft* hose (ID= 1.50 *cm*) as shown in the top of Figure 1.50. The height h is 1.8 *m*. Calculate the average velocity of water in the hose. Do not neglect friction; you may assume turbulent flow.



PROBLEM: 3.42 An incompressible fluid enters a converging bend flowing at a steady flow rate of Q gpm (Figure 3.42). The bend makes a 20° turn and has a circular cross-section all along its length. At the inlet to the bend the radius of the conduit is R_1 and at the exit the radius is a smaller value, R_2. What is the average velocity of the fluid entering the conduit in terms of these variables? What is the average velocity of the fluid exiting the conduit?

PROBLEM: 6.41 Flow Problem: Combined backward pressure and forward drag of a Newtonian fluid in a slit. Calculate the velocity profile for the flow shown in Figure 6.21. The flow is steady flow of an incompressible, Newtonian fluid between two wide plates. The flow is driven forward by the motion of the top plate (the top plate moves in the x-direction at speed V), and the flow is opposed by the pressure, which is slightly higher at the exit,P_L, than at the entrance, P_0, $P_L > P_0$. You may neglect the effect of gravity. Use the coordinate system given in Figure 6.21.



PROBLEM: 9.19 What is the direction and magnitude of the force needed to support the 90° expanding pipe bend shown in Figure 9.47? The water flow is steady and turbulent, and the cross-section of the inlet of the pipe bend is πR_1^2 and the cross-section of the outlet of the bend is πR_2^2, where $R_2 > R_1$. Evaluate your answer for $R_1 = 0.545\ in$ and $R_2 = 0.834\ in$ and various values of flow parameters.

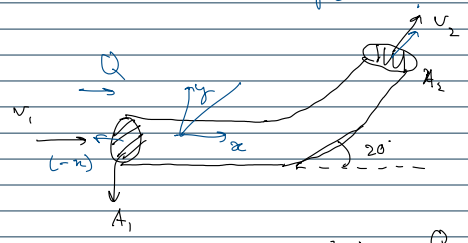
Week of 5th Dec.

Tuesday, December 5, 2023 10:43 AM

PROBLEM: 3.42 An incompressible fluid enters a converging bend flowing at a steady flow rate of Q gpm, (Figure 3.42). The bend makes a 20° turn and has a circular cross-section all along its length. At the inlet to the bend the radius of the conduit is R_1 and at the exit the radius is a smaller value R_2. What is the average velocity of the fluid entering the conduit in terms of these variables? What is the average velocity of the fluid exiting the conduit?

1) Draw diagram
2) Set up the info (w/ the coordinates)
3) knowns, unknowns

How can you solve?



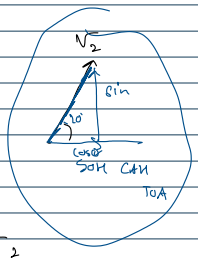
$Q_1 = Q_2$ ✓

$\langle V_1 \rangle = \frac{1}{A_1}\langle V_2 \rangle$ ✓

Q ✓, ρ ✓, A_1

$\langle V_1 \rangle = \frac{Q}{A_1}$

$\langle V_1 \rangle = \frac{Q}{\pi R_1^2}$; $\langle V_2 \rangle = \frac{Q}{\pi R_2^2}$



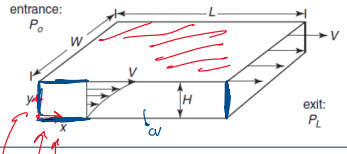
$\rho \cos\theta \langle V \rangle^2 \hat{e}_V$

$\hat{e}_{V_1} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}_{xy}$

$\hat{e}_{V_2} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}_{xy}$

dt, ρ, Navier ✓

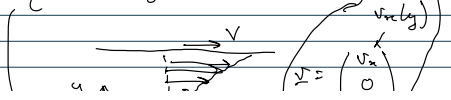
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1 ✓

1) Mass balance ✓ Cartesian x, y

$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$



$\vec{V} = \begin{pmatrix} v_x \\ 0 \\ 0 \end{pmatrix}$

Equation of Continuity (Rectangular Coordinates)

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

Equation of Continuity (Cylindrical Coordinates)

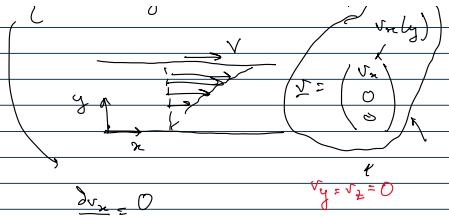
$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

Equation of Continuity (Spherical Coordinates)

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}(\rho r^2 v_r) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \theta}(\rho v_\theta \sin\theta) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \phi}(\rho v_\phi) = 0$$

Cartesian coordinates:

$\rho\left(\frac{\partial v_x}{\partial t} + v_x\frac{\partial v_x}{\partial x} + v_y\frac{\partial v_x}{\partial y} + v_z\frac{\partial v_x}{\partial z}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2}\right) + \rho g_x$ B.7-1



$\frac{\partial v_x}{\partial x} = 0$ (Developed flow) ✓

Cartesian coordinates:

$$\rho\left(\frac{\partial v_x}{\partial t}+v_x\frac{\partial v_x}{\partial x}+v_y\frac{\partial v_x}{\partial y}+v_z\frac{\partial v_x}{\partial z}\right)=-\frac{\partial p}{\partial x}+\mu\left(\frac{\partial^2 v_x}{\partial x^2}+\frac{\partial^2 v_x}{\partial y^2}+\frac{\partial^2 v_x}{\partial z^2}\right)+\rho g_x \quad \text{B.7-1}$$

$$\rho\left(\frac{\partial v_y}{\partial t}+v_x\frac{\partial v_y}{\partial x}+v_y\frac{\partial v_y}{\partial y}+v_z\frac{\partial v_y}{\partial z}\right)=-\frac{\partial p}{\partial y}+\mu\left(\frac{\partial^2 v_y}{\partial x^2}+\frac{\partial^2 v_y}{\partial y^2}+\frac{\partial^2 v_y}{\partial z^2}\right)+\rho g_y \quad \text{B.7-2}$$

$$\rho\left(\frac{\partial v_z}{\partial t}+v_x\frac{\partial v_z}{\partial x}+v_y\frac{\partial v_z}{\partial y}+v_z\frac{\partial v_z}{\partial z}\right)=-\frac{\partial p}{\partial z}+\mu\left(\frac{\partial^2 v_z}{\partial x^2}+\frac{\partial^2 v_z}{\partial y^2}+\frac{\partial^2 v_z}{\partial z^2}\right)+\rho g_z \quad \text{B.7-3}$$

(2) Navier stokes:

x: $0 = -\frac{\partial p}{\partial x} + \mu\frac{\partial^2 v_x}{\partial y^2}$

$0 = -\frac{\partial p}{\partial y}$

$0 = -\frac{\partial p}{\partial z}$

P is not changing w/ y & z → $P \neq f(y,z)$

$\frac{\partial p}{\partial x} = \mu\frac{\partial^2 v_x}{\partial y^2} = \lambda$

P has no y depen; v_x has no x depen

$\frac{\partial p}{\partial x} = \lambda$

$\frac{\partial^2 v_x}{\partial y^2} = \frac{\lambda}{\mu}$

BC's: v_x: @ $y=0$, $v_x=0$; @ $y=h$, $v_x=V$. P: @ $x=0$, $P=P_0$; @ $x=L$, $P=P_L$

2nd order

$\frac{\partial^2 v_x}{\partial y^2} = \frac{\lambda}{\mu}$

$\frac{\partial v_x}{\partial y} = \phi$

$\frac{\partial \phi}{\partial y} =$

Integrate ↓

$\frac{\partial v_x}{\partial y} = \frac{\lambda}{\mu}y + C_1$

Integrate ↓

$v_x = \frac{\lambda}{2\mu}y^2 + C_1 y + C_2$

@ $y=0$, $v_x=0 \rightarrow C_2=0$

@ $y=h$, $v_x=V \rightarrow V = \frac{\lambda}{2\mu}h^2 + C_1 h$

$\rightarrow C_1 = \left(V - \frac{\lambda h^2}{2\mu}\right)\frac{1}{h} = \left(\frac{V}{h} - \frac{\lambda h}{2\mu}\right)$

$$\boxed{v_x = \frac{\lambda}{2\mu}y^2 + \left(\frac{V}{h} - \frac{\lambda h}{2\mu}\right)y}$$

↳ input λ

1st order

$\frac{\partial P}{\partial x} = \frac{dP}{dx} = \lambda$

$P = \lambda x + C$

@ $x=0$, $C = P_0$

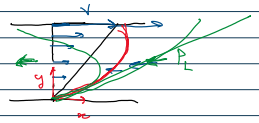
$P = \lambda x + P_0$

@ $x=L$, $P_L = \lambda L + P_0$

$$\boxed{\lambda = \frac{P_L - P_0}{L}}$$ ✓✓

$V = V_{max} \Rightarrow$??

@ max v → $\frac{\partial v_x}{\partial y} = 0$

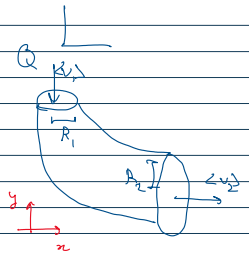


$\frac{\partial}{\partial y}(m) = \frac{\lambda}{\mu}y + \left(\frac{V}{h} - \frac{\lambda h}{2\mu}\right) = 0$

$y = \left(\frac{\lambda h}{2\mu} - \frac{V}{h}\right)\cdot\frac{\mu}{\lambda}$

$$\boxed{y = \frac{h}{2} - \frac{V\mu}{\lambda h}} \rightarrow \neq h$$

PROBLEM: 9.19 What is the direction and magnitude of the force needed to support the 90° expanding pipe bend shown in Figure 9.47? The water flow is steady and turbulent, and the cross-section of the inlet of the pipe bend is πR_1^2 and the cross-section of the outlet of the bend is πR_2^2, where $R_2 > R_1$. Evaluate your answer for $R_1 = 0.545\ in$ and $R_2 = 0.834\ in$ and various values of flow parameters.



(1) Macro

(2) force → momentum → velocity → Mass ($Q_1 = Q_2$)

$\langle V_2 \rangle$, A

$$\frac{d\vec{P}}{dt} = \rho_1 A_1 \cos\theta \langle v_1 \rangle^2 \hat{e}_{v_1} + \rho_2 A_2 \cos\theta \langle v_2 \rangle^2 \hat{e}_{v_2} = \sum f_{on,CV} \quad \begin{pmatrix} 0 \\ 0 \\ -M_{cv} g \end{pmatrix}_{xyz}$$

$$\vec{f}_{in} + \vec{f}_{out} + \vec{f}_g + \vec{f}_\tau$$

$$\vec{f} = \iint_S (\hat{n} \cdot \bar{\bar{\pi}})_{escapes} \, dS$$

$$\bar{\bar{\pi}} = -p\bar{\bar{I}} + \bar{\bar{\tau}}^{\,0}$$

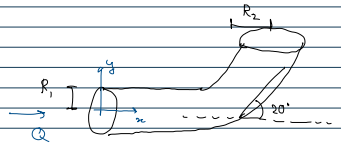
$$\vec{f}_{in} = -P_1 A_1 \hat{n}_1$$

$$\vec{f}_{out} = -P_2 A_2 \hat{n}_2$$

Week of 5th Dec.

Tuesday, December 5, 2023 10:43 AM

PROBLEM: 3.42 An incompressible fluid enters a converging bend flowing at a steady flow rate of Q gpm (Figure 3.42). The bend makes a 20° turn and has a circular cross-section all along its length. At the inlet to the bend the radius of the conduit is R_1 and at the exit the radius is a smaller value, R_2. What is the average velocity of the fluid entering the conduit in terms of these variables? What is the average velocity of the fluid exiting the conduit?



1. Draw ✓
2. Coordinate ✓
3. Balances / equations

$Q = Q_1 = Q_2$ ✓

$\langle v_1 \rangle \cdot A_1 = \langle v_2 \rangle A_2$

$Q = \langle v \rangle \cdot A$

$\langle v_1 \rangle = \frac{Q}{A_1} = \frac{Q}{\pi R_1^2}$ ✓

$\langle v_2 \rangle = \frac{Q}{A_2} = \frac{Q}{\pi R_2^2}$ ✓

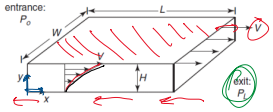
$\langle v \rangle = \frac{\iint v \cdot dA}{\iint dA}$

$\underline{v} = (ax + bx^2 + c)\,\underline{e}_x$

Cauchy → τ → $\tau = \mu \dot{\gamma}^{n-1}$

Navier stokes

PROBLEM: 6.41 Flow Problem: Combined backward pressure and forward drag of a Newtonian fluid in a slit. Calculate the velocity profile for the flow shown in Figure 6.21. The flow is steady flow of an incompressible Newtonian fluid between two wide plates. The flow is driven forward by the motion of the top plate (the top plate moves in the x-direction at speed V), and the flow is opposed by the pressure, which is slightly higher at the exit, P_L, than at the entrance, P_0, $P_L > P_0$. You may neglect the effect of gravity. Use the coordinate system given in Figure 6.21.



1. Draw / study diagram
2. Coordinates ✓
3. Cont & flow Navier ✓
   - v, p

Equation of Continuity (Rectangular Coordinates)

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

Equation of Continuity (Cylindrical Coordinates)

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

Equation of Continuity (Spherical Coordinates)

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}(\rho r^2 v_r) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \theta}(\rho v_\theta \sin\theta) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \phi}(\rho v_\phi) = 0$$

$v_y = v_z = 0$

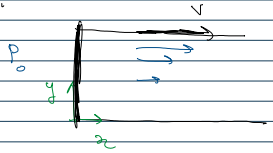
Cartesian coordinates

$$\rho\left(\frac{\partial v_x}{\partial t} + v_x\frac{\partial v_x}{\partial x} + v_y\frac{\partial v_x}{\partial y} + v_z\frac{\partial v_x}{\partial z}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2}\right) + \rho g_x \quad \text{B.7-1}$$

$$\rho\left(\frac{\partial v_y}{\partial t} + v_x\frac{\partial v_y}{\partial x} + v_y\frac{\partial v_y}{\partial y} + v_z\frac{\partial v_y}{\partial z}\right) = -\frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2}\right) + \rho g_y \quad \text{B.7-2}$$

$$\rho\left(\frac{\partial v_z}{\partial t} + v_x\frac{\partial v_z}{\partial x} + v_y\frac{\partial v_z}{\partial y} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial p}{\partial z} + \mu\left(\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2}\right) + \rho g_z \quad \text{B.7-3}$$

velocity ??



$\underline{v} = \begin{pmatrix} v_x \\ 0 \\ 0 \end{pmatrix}$ $\quad v_x = f(y)$

$v_y = v_z = 0$

(1) Mass: $\nabla \cdot \underline{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0 \rightarrow \frac{\partial v_x}{\partial x} = 0$ ✓ (Developed flow)

(2) Navier: $-\frac{\partial p}{\partial x} + \mu \frac{\partial^2 v_x}{\partial y^2} = 0$

PROBLEM: 9.19 What is the direction and magnitude of the force needed to support the 90° expanding pipe bend shown in Figure 9.47? The water flow is steady and turbulent, and the cross-section of the inlet of the pipe bend is πR_1^2 and the cross-section of the outlet of the bend is πR_2^2, where $R_2 > R_1$. Evaluate your answer for $R_1 = 0.545$ *in* and $R_2 = 0.834$ *in* and various values of flow parameters.